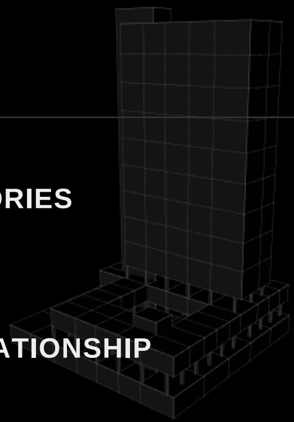
LEVER HOUSE

Building Graph Representation

PHASE 03 | GRAPH CLASSIFICATION



// 01 **CELL COMPLEX**
> massing decomposition

// 02 **CELL CATEGORIES**
> element taxonomy

// 03 **ADJACENCY GRAPH**
> topological connectivity

// 04 **GROUND RELATIONSHIP**
> classification result

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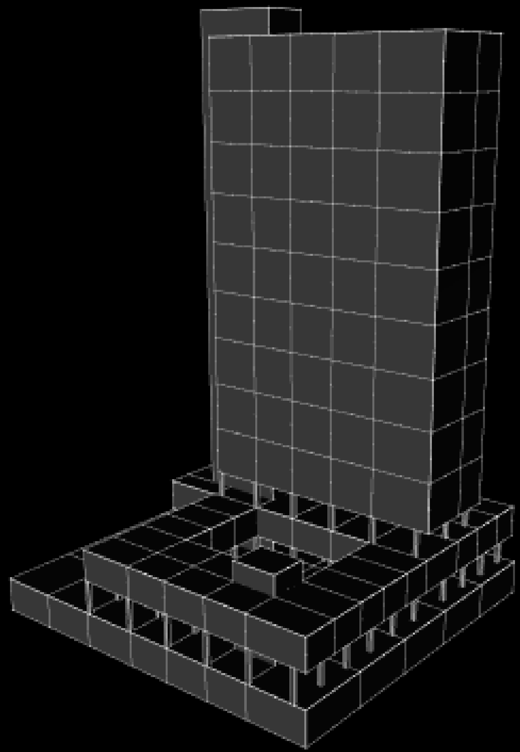
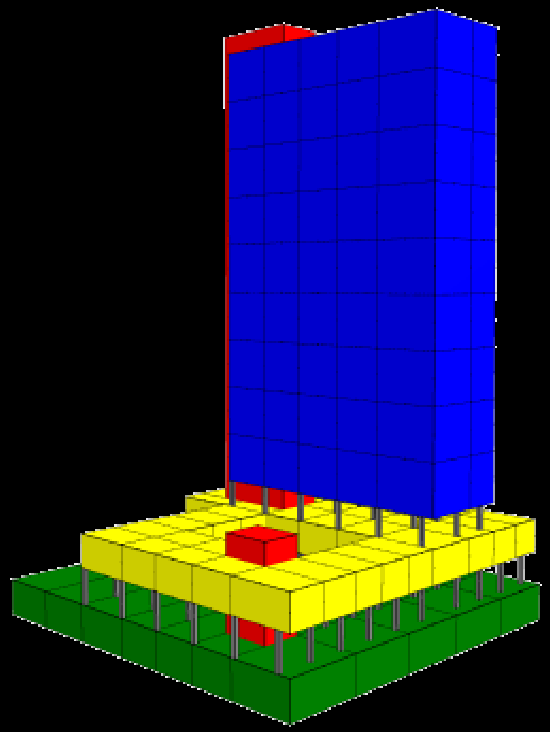


// 01

CELL COMPLEX

| MASSING

> tower, podium, plinth and core merged into one cell complex
> shared faces between adjacent volumes are detected automatically
> the building mass becomes a set of connected topological cells



// 02

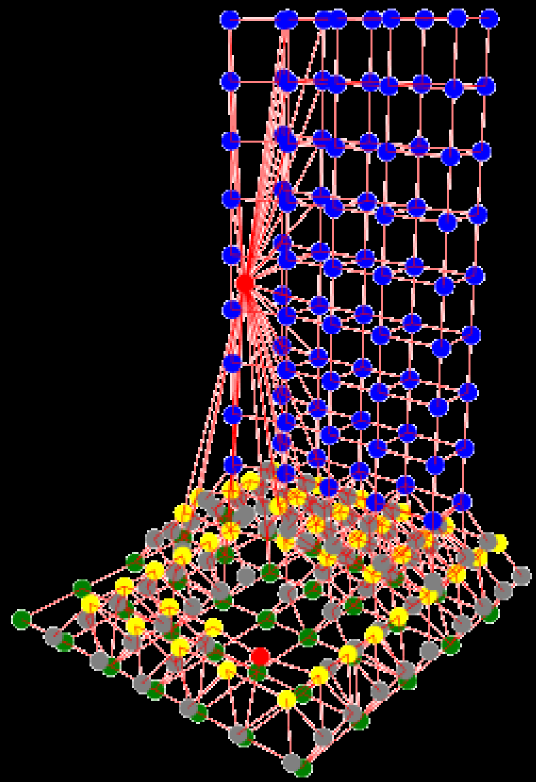
CELL CATEGORIES

| TAXONOMY

> every cell tagged by element type: ground, column, plinth, office, core
> categories one-hot encoded into five-dimensional node features
> colour encodes the structural role of each volume

ELEMENT TYPE

- Ground
- Plinth
- Core
- Columns
- Offices



// 03

ADJACENCY GRAPH

| CONNECTIVITY

> each cell becomes a node; shared faces become edges
> node colour is inherited from the element category
> the building is now a graph the network can read

GRAPH

Node = cell

Edge = shared face

	Actual Value	Predicted Value	Actual Label	Predicted Label	Confidence
0	1	1	Separation with Plinth	Separation with Plinth	1.0

// 04

GROUND RELATIONSHIP

| CLASSIFICATION

> GraphSAGE classifier trained on labelled massing typologies
> Lever House predicted as "Separation with Plinth"
> prediction matches the assigned label at 1.00 confidence

PREDICTED CLASS

0 Separation
1 Separation with Plinth
2 Adherence
3 Adherence with Plinth
4 Interlock